

Linear Algebra II (2026 Fall)

Lecture 2.

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Chapter 5. Diagonalization

Recall : Let V be an n -dimensional vector space over a field F and let

$$T : V \rightarrow V$$

be a linear transformation (i.e. T is a linear operator on V).

(1) Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis for V . Then for each $1 \leq i \leq n$,

$$T(v_i) = a_{1i}v_1 + a_{2i}v_2 + \dots + a_{ni}v_n \text{ for some } a_{1i}, a_{2i}, \dots, a_{ni} \in F.$$

We call $[T(v_i)]_{\mathcal{B}} = \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{pmatrix}$ the coordinate vector of $T(v_i)$ relative to $\mathcal{B}$.

Also the $n \times n$ square matrix

$$[T]_{\mathcal{B}} = \begin{bmatrix} [T(v_1)]_{\mathcal{B}} & [T(v_2)]_{\mathcal{B}} & \cdots & [T(v_n)]_{\mathcal{B}} \end{bmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

is called the matrix representation of T in the ordered basis $\mathcal{B}$.

This chapter is devoted to the so-called **diagonalization problem**.

Let us begin by briefly explaining what this problem is.

Remark : Let F be a field and let $A \in M_{n \times n}(F)$ be an $n \times n$ square matrix.

Consider the linear transformation $L_A : F^n \rightarrow F^n$, $v \mapsto Av$.

(1) Let $\mathcal{B} = \{u_1, \dots, u_n\}$ be a basis for F^n .

If $v = a_1u_1 + \dots + a_nu_n$, then

$$L_A(v) = Av = A(a_1u_1 + \dots + a_nu_n) = a_1Au_1 + \dots + a_nAu_n.$$

Thus

$$R(L_A) = \text{span}(\{Au_1, Au_2, \dots, Au_n\})$$

and

L_A is uniquely determined by $Au_1, Au_2, \dots, Au_n$.

Remark : (Continued) Let F be a field and let $A \in M_{n \times n}(F)$ be an $n \times n$ square matrix.

(2) Let $A = [v_1 \ v_2 \ \cdots \ v_n]$ where $v_1, \dots, v_n \in F^n$ are column vectors of A .

Also let $\{e_1, \dots, e_n\}$ be the standard basis for F^n .

Then $L_A(e_i) = Ae_i = v_i$.

Therefore if $v = (a_1, \dots, a_n) = a_1e_1 + \cdots + a_ne_n$, then

$$L_A(v) = Av = A(a_1e_1 + \cdots + a_ne_n) = a_1Ae_1 + \cdots + a_nAe_n = a_1v_1 + \cdots + a_nv_n.$$

(3) Suppose that A is a diagonal matrix. That is,

$$A = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in F.$$

Equivalently, $v_i = \lambda_i e_i$ for all $1 \leq i \leq n$.

Then $L_A(e_i) = Ae_i = \lambda_i e_i$ for all $1 \leq i \leq n$, and hence

$$L_A(a_1e_1 + \cdots + a_ne_n) = a_1\lambda_1e_1 + \cdots + a_n\lambda_ne_n.$$

Thus if A is a diagonal matrix, then

the corresponding linear transformation $L_A : F^n \rightarrow F^n$ has a very simple form.

Remark : (Continued) Let F be a field and let $A \in M_{n \times n}(F)$ be an $n \times n$ square matrix.

(4) Let us generalize the case where A is a diagonal matrix as follows.

Suppose that there exists an ordered basis $\mathcal{B} = \{u_1, \dots, u_n\}$ for F^n such that

$$L_A(u_i) = Au_i \text{ is equal to } \lambda_i u_i \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in F.$$

Then we have

$$L_A(a_1 u_1 + \dots + a_n u_n) = A(a_1 u_1 + \dots + a_n u_n) = a_1 A u_1 + \dots + a_n A u_n = a_1 \lambda_1 u_1 + \dots + a_n \lambda_n u_n.$$

Also

$$\begin{aligned} [L_A]_{\mathcal{B}} &= \begin{bmatrix} [Au_1]_{\mathcal{B}} & [Au_2]_{\mathcal{B}} & \cdots & [Au_n]_{\mathcal{B}} \end{bmatrix} \\ &= \begin{bmatrix} [\lambda_1 u_1]_{\mathcal{B}} & [\lambda_2 u_2]_{\mathcal{B}} & \cdots & [\lambda_n u_n]_{\mathcal{B}} \end{bmatrix} \\ &= \begin{bmatrix} \lambda_1 e_1 & \lambda_2 e_2 & \cdots & \lambda_n e_n \end{bmatrix} \\ &= \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}. \end{aligned}$$

That is, $[L_A]_{\mathcal{B}}$ is a diagonal matrix.

Let $T : V \rightarrow V$ be a linear transformation from an n -dimensional vector space V to itself. This chapter is concerned with the so-called **diagonalization problem** of $T : V \rightarrow V$. That is, we seek answers to the following two questions.

Q1. : Does there exist an ordered basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}$ is a diagonal matrix?

Q2. : If such a basis exists, how can it be found?

Remark : If $[T]_{\mathcal{B}}$ is a diagonal matrix, then $T(v_i) = a_{ii}v_i$ for all $1 \leq i \leq n$.

Therefore for any $v = b_1v_1 + b_2v_2 + \cdots + b_nv_n \in V$, it holds that

$$T(v) = b_1T(v_1) + b_2T(v_2) + \cdots + b_nT(v_n) = b_1a_{11}v_1 + b_2a_{22}v_2 + \cdots + b_na_{nn}v_n.$$

Thus if $[T]_{\mathcal{B}}$ is a diagonal matrix, then

we can understand more clearly how T acts on V .

Therefore if there exists a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}$ is a diagonal matrix, then we can expect to obtain easy solutions to

many practical problems that can be formulated in a linear algebra context.

Example : Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the reflection along the line $y = 2x$.

(1) T is a linear transformation.

(2) Let $u = (1, 2)$ be a point in the line $y = 2x$. Then $T(u) = u$.

(3) Note that the line $y = -\frac{1}{2}x$ is perpendicular to the line $y = 2x$.

Now, let $v = (2, -1)$ be a point in the line $y = -\frac{1}{2}x$. Then $T(v) = -v$.

(4) Since $\beta = \{u, v\}$ is a basis of $\mathbb{R}^2$, it holds that

$$T(au + bv) = au - bv \text{ for all } a, b \in \mathbb{R}.$$

(5) $[T]_{\beta} = \begin{bmatrix} [u]_{\beta} & [-v]_{\beta} \end{bmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is a **diagonal matrix**.

Section 5.1 Eigenvalues and Eigenvectors

Recall : Let V be an n -dimensional vector space over a field $\mathbb{F}$ and let

$$T : V \rightarrow V$$

be a linear transformation. The main goal of this chapter is

to find an ordered basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}$ is a diagonal matrix.

In other words, it needs to find a basis $\mathcal{B} = \{v_1, \dots, v_n\}$ for V such that

$$T(v_i) = c_i v_i \text{ for some } c_i \in \mathbb{F}.$$

Example : Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the reflection of $\mathbb{R}^2$ about the line $y = 2x$. Then one can easily see that the two vectors $v_1 = (1, 2)$ and $v_2 = (-2, 1)$ satisfy the equality

$$T(v_1) = v_1 \text{ and } T(v_2) = -v_2.$$

Thus the basis $\{v_1, v_2\}$ for $\mathbb{R}^2$ satisfies the desired property that $[T]_{\mathcal{B}}$ is a diagonal matrix.

More precisely,

$$[T]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Definition 1. : Let V be an n -dimensional vector space over a field $\mathbb{F}$ and let

$$T : V \rightarrow V$$

be a linear transformation. We say that T is **diagonalizable** if

there exist an ordered basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}$ is a diagonal matrix.

Definition 2. : Let A be an $n \times n$ square matrix. Then we say that A is **diagonalizable** if

the linear transformation $L_A : \mathbb{F}^n \rightarrow \mathbb{F}^n$ is diagonalizable.

Remark : Suppose that the $n \times n$ square matrix A is diagonalizable.

Let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be a basis for $\mathbb{F}^n$ such that $[L_A]_{\mathcal{B}}$ is a diagonal matrix.

Thus $L_A(v_i) = Av_i = c_i v_i$ for all $1 \leq i \leq n$. Then

$$A[v_1 \ v_2 \ \cdots \ v_n] = [c_1 v_1 \ c_2 v_2 \ \cdots \ c_n v_n] = [v_1 \ v_2 \ \cdots \ v_n]D \text{ where } D = \begin{pmatrix} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n \end{pmatrix}.$$

Also the square matrix $Q := [v_1 \ v_2 \ \cdots \ v_n]$ is invertible since $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ is a basis.

Thus we have $AQ = QD$ and hence $A = QDQ^{-1}$.

Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

We want to determine

when T is diagonalizable

and, if so,

how to obtain an ordered basis $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ such that $[T]_{\mathcal{B}}$ is a diagonal matrix.

(1) If $D = [T]_{\mathcal{B}}$ is a diagonal matrix, then $T(v_i) = \lambda_i v_i$ for all $1 \leq i \leq n$.

(2) Conversely, if $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ is an ordered basis for V such that

$$T(v_i) = \lambda_i v_i \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}.$$

Then

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

In the preceding slide, each vector v in the basis $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ satisfies the condition

$$T(v) = \lambda v \text{ for some scalar } \lambda \in \mathbb{F}.$$

Moreover, $v \neq 0_V$, since v is a member of the basis $\mathcal{B}$.

These observations motivate the following definitions.

Definition 3. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V .

A nonzero vector $v \in V$ is called **an eigenvector of** T if there exists a scalar $\lambda \in \mathbb{F}$ such that

$$T(v) = \lambda v.$$

The scalar λ is called **the eigenvalue** corresponding to the eigenvector v .

Definition 4. : Let $A \in M_{n \times n}(\mathbb{F})$ be a square matrix.

A nonzero vector $v \in \mathbb{F}^n$ is called **an eigenvector of** A if it is an eigenvector of L_A .

In this case, $Av = \lambda v$ for a scalar $\lambda \in \mathbb{F}$.

The scalar λ is called **the eigenvalue** of A corresponding to the eigenvector v .

※ **(Terminologies)** eigenvector = characteristic vector = proper vector

Theorem 5.1. : Let $T : V \rightarrow V$ be a linear transformation of an n -dimensional vector space V . Then T is diagonalizable if and only if

there exists an ordered basis $\mathcal{B}$ for V consisting of eigenvectors of T .

Furthermore, if T is diagonalizable and $\mathcal{B} = \{v_1, \dots, v_n\}$ is an ordered basis of eigenvectors of T , then

$D := [T]_{\mathcal{B}}$ is a diagonal matrix

and

D_{jj} is the eigenvalue corresponding to v_j for $1 \leq j \leq n$.

Remark : Theorem 5.1 shows that

diagonalizing a matrix or a linear operator

is equivalent to

finding a basis of eigenvectors and the corresponding eigenvalues.

Example 1. : Let

$$A = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}, v_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ and } v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

$$(1) L_A(v_1) = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} = -2 \times \begin{pmatrix} 1 \\ -1 \end{pmatrix} = -2v_1. \text{ Thus}$$

v_1 is an eigenvector of L_A (and so of A), and $\lambda_1 = -2$ is the eigenvalue corresponding to v_1 .

$$(2) L_A(v_2) = \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 15 \\ 20 \end{pmatrix} = 5 \times \begin{pmatrix} 3 \\ 4 \end{pmatrix} = 5v_2. \text{ Thus}$$

v_2 is an eigenvector of L_A (and so of A), and $\lambda_2 = 5$ is the eigenvalue corresponding to v_2 .

(3) $\mathcal{B} = \{v_1, v_2\}$ is an ordered basis for $\mathbb{R}^2$ consisting of eigenvectors of both A and L_A .

Therefore A and L_A are diagonalizable.

Moreover, by Theorem 5.1, it holds that $[L_A]_{\mathcal{B}} = \begin{pmatrix} -2 & 0 \\ 0 & 5 \end{pmatrix}$.

Example 2. : Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the rotation through an angle of $\pi/2$.

It is clear geometrically that no nonzero vector can be an eigenvector of T .

Thus there exist operators (and matrices) with no eigenvalues or eigenvectors.

In consequence, there are operators and matrices which are not diagonalizable.

Next Problem : How to find eigenvectors and eigenvalues of a given matrix $A \in M_{n \times n}(\mathbb{F})$?

Theorem 5.2. : Let $A \in M_{n \times n}(\mathbb{F})$ and let $\lambda \in \mathbb{F}$. Then

λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Proof : λ is an eigenvalue of $A \Leftrightarrow$ there exists a nonzero vector $v \in \mathbb{F}^n$ such that $Av = \lambda v$

$$(Av = \lambda v \Leftrightarrow Av = \lambda I_n v \Leftrightarrow (A - \lambda I_n)v = \mathbb{O})$$

$$\Leftrightarrow N(A - \lambda I_n) \text{ is not the zero space and so } \text{rank}(A - \lambda I_n) < n$$

$$\Leftrightarrow A - \lambda I_n \text{ is not invertible}$$

$$\Leftrightarrow \det(A - \lambda I_n) = 0. \quad \blacksquare$$

Definition : For $A \in M_{n \times n}(\mathbb{F})$,

the polynomial $f(t) := \det(A - tI_n)$ is called the **characteristic polynomial** of A .

Remark : Theorem 5.2 states that

the eigenvalues of A are the zeros of its characteristic polynomial $\det(A - tI_n)$.